

Winter Examination - 2022

Branch: - Common for All branches

Semester: - III

Course: - B Tech

Subject Code & Name: BTBS301

Engineering Mathematics-III

Max. Marks: - 60

Date: - 09/03/2023

Duration: - 3-Hrs

Instructions to the Students:

- All the questions are compulsory.
- The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
- Use of non-programmable scientific calculators is allowed.
- Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q.1 Solve Any Three of the following. 12

A) Find Laplace Transform of $e^{-3t} \sin^2 t$ L3/CO1 4

B) Find Laplace Transform of $f(t) = \begin{cases} 1 & 0 < t < 1 \\ 0 & 1 < t < 2 \end{cases}$ L3/CO1 4

where $f(t)$ is periodic function of period 2.

C) Evaluate using Laplace Transform.: $\int_0^\infty \frac{\cos 4t - \cos 3t}{t} dt$ L3/CO1 4

D) Find Laplace Transform of $(1 + 2t - 3t^2 + 4t^3)H(t - 2)$ L3/CO1 4

Q2 Solve Any Three of the following. 12

A) Find the inverse Laplace transformation of the function. $\log \left(1 + \frac{a^2}{s^2} \right)$ L3/CO2 4

B) By using convolution theorem find $L^{-1} \left[\frac{s}{(s^2+4)(s^2+9)} \right]$ L3/CO2 4

C) Find the inverse Laplace transformation of the function. $\frac{5s^2-15s-11}{(s+1)(s-2)^2}$ L3/CO2 4

D) Solve using Laplace transformation

$y'' + 3y' + 2y = t\delta(t - 1)$ for which $y(0) = y'(0) = 0$ L3/CO2 4

L3/CO3

Q.3 Solve Any Three of the following.

A) Using Parseval's identity prove that $\int_0^\infty \frac{x^2}{(x^2+1)^2} dx = \frac{\pi}{4}$

L3/CO3

B) Find the Fourier transform of

$$f(x) = \begin{cases} 1-x^2, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases}$$

L3/CO3

C) Find the Fourier Sine transform e^{-ax} , $a > 0$ D) Find the Fourier cosine transform of the function $f(y) = \begin{cases} \cos y, & 0 < y < a \\ 0, & y > a \end{cases}$

L3/CO3

(12)

Q.4 Solve Any Three of the following.

A) Form the partial differential equation by eliminating arbitrary constants from

L3/CO4

$$(x-a)^2 + (y-b)^2 = z^2 \cot^2 \alpha$$

B) Solve the Partial differential equation $x(y-z)p + y(z-x)q = z(x-y)$

L3/CO4

C) Use the method of separation of variables to solve

$$\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u \quad \text{given that } u(x, 0) = 6e^{-3x}$$

L3/CO4

D) A bar with insulated at its ends is initially at temperature 0°C throughout. The end $x = 0$ is at 0°C for all times and the heat is suddenly applied so that $\frac{\partial u}{\partial x} = 10$ at $x = t$ for all time. Find temperature function $u(x, t)$

L3/CO4

Q.5. Solve Any Three of the following.

(12)

A) Determine k such that the function $f(z) = e^x \cos y + ie^x \sin ky$ is analytic. L3/CO5B) Show that $u = x^2 - y^2 - 2xy - 2x + 3y$ is a harmonic function and hence determine the analytic function $f(z)$ in terms of z . L3/CO5C) Determine the pole of the function $f(z) = \frac{2z-1}{z(z+1)(z-3)}$ and also find the residue at each pole & sum of all residues.

D) Evaluate

$$\oint_C \frac{\sin \pi z^2 + 2z}{(z-1)^2(z-2)} dz \quad \text{Where } C \text{ is the circle } |z| = 1$$

L3/CO5

L3/CO5

*** End ***